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Research on Image Processing Based on Improved Particle Swarm Optimization

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***Abstract*-In recent years, with the rapid development of artificial intelligence technology and computer technology, the combination of graphic technology and mathematics has become increasingly close, which greatly promote the development of digital image processing technology. Because of the differences in the task of image processing, the corresponding processing procedures and models are also different, and there is no uniform evaluation criteria for image processing evaluation, therefore, it needs to set the optimal target, so it is necessary to introduce the improved particle swarm algorithm for image processing, so as to obtain the best image processing results. Based on the analysis of the research status of image processing technology, we set up the corresponding model based on improved particle swarm algorithm to process the image and the simple image optimization process. And we carry out the initial selection of the optimization algorithm, the establishment of the optimization target, and the corresponding optimization of the mathematical model. The final result proves that this is a more effective image processing method.**

Keywords: image processing; particle swarm optimization; dynamic optimization

I INTRODUCTION

With the development of science and technology, the development of computer technology continues to progress, its application is also deepening, human beings have entered the information age. Human transmission of information in the main carrier and the media is the sound and the image, and the image is the main carrier of information transmission, according to research and statistics show that the information obtained from the image of the total information is 70 ~ 80% [1 ~ 2], so the study of the image is very important. Due to the continuous development of multimedia technology and access to the image of the increasing channels, the application of image technology is becoming more and more popular. The use of binoculars, cameras, microscopes and other equipment can make people observe things that cannot be seen with the naked eye , such as microorganisms, cells, the universe, etc., so that people's vision has been greatly expanded. However, in the process of acquiring an image, the image quality of the acquired image is degraded due to noise factors such as the light intensity and the device structure due to the external environment and its own equipment. It is necessary to suppress the noise of the acquired image to improve the image quality.

At present, most of the digital image system, the input image is the firstly freeze and then scan the multi-dimensional image into a one-dimensional signal, and then its processing, storage, transmission and other processing transform. And finally it also will be composed of multi-dimensional image signal, and image noise will also be subject to such decomposition and synthesis. In these processes, the electrical system and the external influence will make the accurate analysis of the image noise become very complicated [3]. On the other hand the image is only the medium of transmission of visual information, the understanding of the image information is understood by the human visual system. Different image noise, people feel the degree is different, this is the so-called human noise visual characteristics of the subject.

The importance of image noise in digital image processing technology is becoming more and more obvious, such as high magnification aerial interpretation, X-ray image system in the noise removal has become an indispensable technical steps.

II THE REASON FOR NOISE GENERATION AND THE BASIC METHOD OF DE-NOISING

A. Based on the cause, it can be divided into external noise and internal noise

External noise, refers to the external interference system to electromagnetic waves or by the power into the system caused by the internal noise. Such as electrical equipment, objects discharge phenomena caused by noise.

Internal noise: it generally can be divided into the following four types:

The noise caused by the basic properties of light and electricity. If the current is generated by the electron or hole particles of the collection, formed by directional movement.

The random noise formed by the randomness of these particles, the thermal noise formed by the irregular thermal motion of free electrons in the conductor, according to the particle properties of light, the image is transmitted by the photon quantum, and light quantum density with time and space changes formed the light quantum noise.

Electrical noise generated by mechanical movement. Such as various joints due to jitter caused by changes in current generated by the noise; head, tape and other jitter or jitter together.

Equipment material itself caused by noise. Such as the surface particles of the positive and negative films and the noise generated by the surface defects of the magnetic disk. With the development of materials science, these noise is expected to continue to decrease, but at present, it is inevitable.

System internal equipment circuit caused by the noise. Such as AC noise introduced by the power supply; noise caused by the deflection system and the clamp circuit.

The effect of noise on the signal can be divided into additive noise and multiplicative noise, but in the digital image processing are generally additive noise, so here

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only consider the impact of additive noise on the image. Let the received signal be r (t), the original signal is s (t),

The noise signal is n (t) independent of s (t) that contains the additive noise signal model can be expressed as [4]:

r(t) = s(t) + n(t)

B. Denoising method

（1）Mean filter

The mean filter using the neighborhood averaging method is well suited for removing the particle noise in the image obtained by scanning. The domain averaging method suppresses the noise effectively, and also causes the blurring phenomenon due to the average, and the fuzzy degree is proportional to the domain radius.

The smoothness achieved by the geometric mean filter can be compared to the arithmetic mean filter, but less image detail is lost during the filtering process.

Harmonic mean filter is better for "salt" noise, but does not apply to "pepper" noise. It is good at dealing with other noise like Gaussian noise.

The inverse harmonic mean filter is more suitable for dealing with impulse noise, but it has a drawback that it is necessary to know whether the noise is dark noise or bright noise in order to select the appropriate filter order symbol, the order of the wrong choice of signs may cause catastrophic consequences.

operation process we do not require the statistical characteristics of the image, which also brings a lot of convenience, but for some details, especially points, lines, steeple more images we should not use the median filter method.

（4）Morphological noise filter

Combined open and closed together which can be used to filter out the noise, first of all the open the operation to noise image, you can choose the structure of the matrix which is larger than the size of the noise, and thus open the result in order to remove the noise on the background. Finally, to close the operation of the image get from the previous step, remove the noise on the image. According to the characteristics of this method we can know that this method applies to the image type that the image of the object size is relatively large, and no small details, for this type of image the noise reduction effect will be better [5].

（5）Wavelet denoising

This method preserves most of the wavelet coefficients that contain the signal, so the image detail can be better preserved [6]. Wavelet analysis for image denoising mainly has three step: (1)The image signal is decomposed by wavelet.(2)The threshold is quantized by the high frequency coefficients after the decomposition.(3)Reconstructing image signals using two - dimensional wavelet.

III Improved Particle Swarm Optimization （IPSO）

A. Basic Particle Swarm Optimization （PSO）

In the international neural network academic conference of IEEE, 1995, American social psychologist J.Kennedy and electrical engineer R.C.Eberhart formally published the article Particle Swarm Optimization, marking the birth of POS[7]. The basic idea is assuming a particle swam containing M particles can fly at a certain speed in D-dimensional search space. The attribute of particle i at t time is as follows.

Position: $x_i^t = (x_{i1}^t, x_{i2}^t, \cdots, x_{id}^t)^T$ (1)

$x_{id}^t \in [L_d, U_d]$, L_d and U_d are respectively upper limit and lower limit of the search space.

Velocity: $v_i^t = (v_{i1}^t, v_{i2}^t, \cdots, v_{id}^t)^T$ (2)

$v_{id}^t \in [v_{\min,d}, v_{\max,d}]$, $v_{\min}$ and $v_{\max}$ are respectively the smallest and largest Velocity.

Optimal position of individuals: $p_i^t = (p_{i1}^t, p_{i2}^t, \cdots, p_{iD}^t)^T$; global Optimal position: $1 \le d \le D, 1 \le i \le M$. $p_g^t = (p_{g1}^t, p_{g2}^t, \cdots, p_{gD}^t)^T$.

Then position of particle at time $t+1$ can be expressed in the following equation:

$$v_{id}^{t+1} = v_{id}^t + c_1 r_1^t (p_{id}^t - x_{id}^t) + c_2 r_2^t (p_{gd}^t - x_{id}^t) \quad (3)$$

$$x_{id}^{t+1} = x_{id}^t + v_{id}^{t+1} \quad (4)$$

c_1, c_2 are learning factors. Appropriate c_1 and c_2 can accelerate convergence and avoid falling into local optimum. r_1 and r_2 are any number between 0 and 1.

Let the minimized objective function be f (x), then the individual optimal position of particle i is:

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$$y_i(t+1)=\begin{cases} y_i(t) & f(x_i(t+1)) \geq f(y_i(t)) \\ x_i(t+1) & f(x_i(t+1)) \geq f(y_i(t)) \end{cases} \quad (5)$$

B. Improved Particle Swarm Optimization

With the increase of dimension of optimization problem, the basic PSO algorithm is prone to be premature, go into stagnation and even local extremum. On this basis, some people put forward many improved algorithms. Many scholars optimized the performance of particle swarm optimization by adding an appropriate inertia weight w, and gained some fruits.In 1999, M.Clerc suggested introducing constriction factor to PSO algorithm to replace traditional inertia weight. The velocity can be presented as:

$$v_{id}^{t+1}=K(v_{id}^{t}+c_1 r_1^t(p_{id}^t-x_{id}^t)+c_2 r_2^t(p_{gd}^t-x_{id}^t)) \quad (6)$$

Constriction factor K:

$$k=\frac{2}{\left|2-\varphi+\sqrt{\varphi^2-4\varphi}\right|}$$

$$\varphi=c_1+c_2, \varphi>4 \quad (7)$$

Compared with inertia weight, constriction factor K can control the velocity of particles more effectively, enhancing the local searching ability of the algorithm.

C. Improved Particle Swarm Simulation Analysis

The standard particle swarm algorithm and the improved particle swarm algorithm are tested with three Benchmark functions.

(1) Rastrigin function

$$f_1(x)=\sum_{i=1}^{n}(x_i^2-10\cos 2\pi x_i)+10), x_i \epsilon [-5.12,5.12] \quad (8)$$

The performance of the algorithm is tested and compared with the traditional particle swarm algorithm. The results are shown in Fig. 1

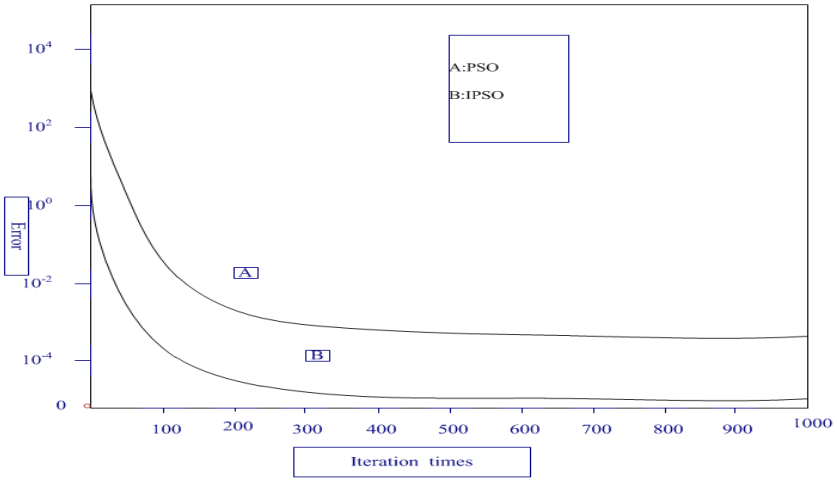


Figure1.Comparison of Iterative Optimization Capability before and after Algorithm Improvement

IV CONCLUSION

In this paper, the denoising problem in image processing is analyzed, and the principle of the most commonly used particle group denoising algorithm is analyzed. And an optimal weight adjustment strategy is proposed to solve the problem of inertia weight distribution in particle swarm noise. The proposed strategy divides the weight distribution into two processes, and different processes apply different adjustment functions.The results show that the proposed optimization algorithm can significantly improve the speed and precision of the search, and the image denoising will be improved obviously.

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